CO_2 Emissions of Large Merchandise Ships

SHS HUM-470
Economic Growth and Sustainability II

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Spring 2024-2025

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1. Introduction

Climate plays a significant role in overall productivity and economic growth. Given its widespread global impact on economic growth, climate change has emerged as a key issue for environmental policymakers at both local and international levels. The primary sectors requiring adaptation and mitigation policies are agriculture, industry, forestry, transportation, and land use. In order to foster economic and social development, adapting to climate change effects is essential, and to effectively address and mitigate this challenge, it is necessary to formulate policies and strategies on an international level. (Abbass et al., 2022)

Among these sectors, transportation – particularly maritime shipping – plays a critical role in both global commerce and greenhouse gas emissions. The shipping industry is the backbone of international trade and has been a major driven of global economic growth, but it is also highly carbon intensive, with CO_2 emissions growing steadily as the shipping industry has grown to meet demand. As a result, the emissions burden created by shipping is substantial and growing, accounting for approximately 3% of global emissions as of 2022, and at current growth rates, shipping could represent some 10% of global greenhouse gas emissions by 2050. (Wells et al., 2018)(T&E, n.d.)

In addition to affecting the atmosphere, shipping emissions also pose serious public health risks. Ships that burn fossil fuels also release pollutants that diminish the life expectancy of populations living near ports. Teresa Bui, the climate policy director at the environmental organization Pacific Environment, highlighted that pollution levels were especially severe during the COVID-19 pandemic when supply chain disruptions led to a buildup of ships outside the Port of Los Angeles, resulting in pollution equivalent to that generated by nearly 100,000 large trucks per day. (Swanson, 2023)

The shipping industry is now in a state of transition, driven by increasingly stricter emissions standards and growing awareness of its environmental footprint. The future of maritime shipping will not be determined solely by the economics of transporting goods across oceans but also by national and international policy measures, changes in geopolitical dynamics, advancements in technology and operations, and certainly by climate change. (Wells et al., 2018)

This report will explore three key questions:

1. How have CO2 emissions for large merchandise ships evolved over time?
2. What are the main factors influencing CO2 emissions?
3. Is there evidence of decoupling between maritime trade and emissions?

2. Existing Studies on Shipping and CO_2 Emissions

In this section, we are going to perform robust research comprehending sources since 2015 – spanning academic journals, IMO and IPCC reports, OECD/ITF analyses, UNCTAD publications, and industry studies – examining how the growth of seaborne trade is linked to rising CO_2 emissions.

2.1. Contribution to Global Emissions

According to UNCTAD, ships carried about 80 % of global trade by volume in 2019, underpinning global supply chains (Deng and Mi, 2023, p. 1). The IMO's Fourth GHG Study found that total shipping emitted approximately 1.076 billion tonnes of GHGs in 2018 – up 9.6% from 977 million tonnes in 2012 – including roughly 1.056 billion tonnes of CO_2, or 2.89% of global CO_2 emissions versus 2.76 % in 2012 (IMO, 2020, p. 29). A 2023 review by Deng & Mi similarly places shipping at "about 3% of total annual anthropogenic CO_2" (Deng and Mi, 2023, p. 1), and notes that international shipping alone emits more CO_2 than all but the top five or six industrialized nations. These figures refer only to direct exhaust CO_2 from vessel engines – excluding upstream fuel production and non-CO_2 gases – which remain the primary focus of regulatory efforts.

2.2. Historical Trends and Decoupling

From 1990 through the mid-2000s, shipping emissions rose almost in lockstep with booming trade. Rapid globalization and expansion of the world fleet led to nearly doubling CO_2 output between 1990 and 2008 (Robinson, 2023).

Emissions peaked in 2008, when seaborne trade and fuel consumption were both high, before the global recession prompted cost-saving measures such as slow steaming, alongside vessel-design and engine-efficiency improvements. Slow steaming alone is estimated to have cut container-ship CO_2 by about 11% since 2008 without new technology (Cariou, 2011). As a result, even as trade grew roughly 4% annually over 2008-2018, total maritime CO_2 fell by about 1% per year on average, yielding 21-29% lower CO_2 per transport work in 2018 versus 2008 (Hong Liang, 2020).

By the late 2010s, however, the "low-hanging fruit" of efficiency had largely been harvested. From 2012 to 2018, demand growth outstripped efficiency gains: total shipping CO_2 rose by nearly 10% (from 978 Mt to 1,076 Mt) despite continued per-unit improvements (IMO, 2020, p. 29; Robinson, 2023). The ICCT noted that "demand for shipping grew twice as quickly as fuel efficiency improved" in that period (ICCT, 2020), and UNCTAD data show that after 2013 CO_2-intensity improvements per ton-mile slowed while trade volumes accelerated (Deng and Mi, 2023). Consequently, absolute emissions growth resumed in the late 2010s, but at a slower rate than before 2008.

2.3. Current Status and Share

International shipping's annual CO_2 emissions, at about 1 Gt in the late 2010s, match those of a top-ten emitting nation and eat into the remaining carbon budget for 1.5–2°C scenarios. IPCC AR6 (2022) explicitly incorporates shipping – alongside aviation – and warns these sectors are growing faster than others, with current commitments "fall[ing] far short" of Paris goals (IPCC, 2022). Since maritime emissions aren't allocated to individual countries under the UNFCCC but regulated by the IMO, their rise presents a distinct mitigation gap (Deng and Mi, 2023, p. 1). If trends continue, shipping could claim a disproportionate share of future allowances: one IMO commissioned study projects its global CO_2 share could jump to 17% by 2050 if other sectors decarbonize more rapidly (ISEMAR, 2020; Deng and Mi, 2023, p. 1). This looming imbalance underscores the need for ambitious policy and technological measures to "bend the curve" of shipping emissions.

2.4. Future Emissions Projections

Under business-as-usual assumptions, shipping emissions are set to keep rising through mid-century. The Fourth IMO GHG Study projects CO_2 in 2050 to be roughly 50% above 2018 levels under a medium-growth scenario (ICCT, 2020), and the Marine Policy analysis foresees a 90 – 130% increase by 2050 relative to 2008 (Robinson, 2023). These "no-new-policy" trajectories imply that gains in carbon intensity would be overwhelmed by growing demand.

Conversely, aggressive deployment of existing technologies and alternative fuels could nearly eliminate CO_2 from shipping by 2035 (International Transport Forum, 2018, p. 11). In low-demand, 1.5°C-consistent pathways (SSP1-1.9), energy use in 2050 is only ~3% above 2018, whereas high-growth (SSP3-7.0) sees a ~44 % rise (Kramel et al., 2024). Yet even broad efficiency improvements "will not be enough" (Hong Liang, 2020) to meet climate targets. Reflecting this, the IMO's 2023 strategy now aims for net-zero GHG by 2050 with interim cuts of 20-30% by 2030 and 70-80% by 2040.

2.5. Methodologies and Uncertainty

The quantification of shipping's CO_2 has itself been an area of study, as different methodologies sometimes yield varying results. The two primary approaches are:

1. Fuel-based ("*top-down*"): estimate total fuel consumed by ships and multiply by CO_2 emission factors.
2. Activity-based ("*bottom-up*"): compute emissions from individual ship activity data (speed, engine power, time at sea).

Each approach has strengths and weaknesses. Top-down methods, used by the IEA and some academic inventories, rely on global bunker fuel sales or energy statistics (Deng and Mi, 2023, p.4). They are quick but can misattribute or omit vessels. Whereas bottom-up AIS-based models

(IMO GISIS, ICCT) provide voyage-level and spatial emissions detail at the cost of extensive processing and operating-profile assumptions. Despite ~20% variation across inventories (Deng and Mi, 2023), all detect a plateau around 2010 followed by rising CO_2. The IMO's Data Collection System now mandates shipboard fuel-use and CO_2 reporting to improve precision, and the Fourth IMO GHG Study estimates 2018 emissions at 740 Mt CO_2 from international voyages and 276 Mt from domestic shipping (~85–90 % international) (ICCT, 2020).

3. Possible Main Drivers of CO_2 Emissions in Shipping

There are several factors that drive CO_2 emissions in shipping:

3.1. Trade Volume (Activity Level)

Shipping CO_2 emissions scale with total transport work, so expanding trade requires more voyages (and fuel) if nothing else changes. The Third IMO GHG Study found that from 1970 to 2007, CO_2 rose almost one-for-one with trade (elasticity ≈ 1), tracking fluctuations like the oil crises (ICCT, 2020). After 2008, efficiency measures weakened this link. UNCTAD's Review confirms that slower GDP and trade growth, alongside vessel efficiencies, have since tempered emissions growth relative to trade volumes (Deng and Mi, 2023, p. 1). Still, surging shipment volumes remain the primary driver of aggregate fuel use and CO_2 output.

3.2. Fuel Type (Carbon Intensity of Fuel)

Ships today predominantly burn carbon-intensive Heavy Fuel Oil (HFO) and Marine Diesel Oil (MDO), which emit roughly 3.1kg CO_2 per kg fuel (≈85–90 g CO_2/MJ) (Deng and Mi, 2023, p. 1). Liquified Natural Gas (LNG), at ≈75g CO_2/MJ, can reduce CO_2 emissions by 15-20% compared to HFO/MDO, but its adoption is limited to newbuild LNG carriers, a few container and cruise ships, and its benefits are eroded by methane slip, which increases CO_2 impacts (ICCT, 2020). In contrast, truly low-carbon fuels (green ammonia, methanol, hydrogen, advanced biofuels) could eliminate or nearly eliminate direct CO_2, though they require substantial scaling and infrastructure development. Meanwhile, operational measures—shore power at berth, wind-assisted propulsion—can immediately lower the voyage "fuel carbon factor" by reducing onboard fuel use. Together, fuel choice and these measures constitute the critical levers for deep decarbonization of shipping.

3.3. Vessel Size & Design Efficiency

Ship characteristics – size, hull form, and engine technology – dictate how efficiently cargo is moved. Larger vessels benefit from economies of scale (e.g., a single 200,000 DWT bulk carrier uses less fuel per tonne of ore than two 100,000 DWT ships), and successive design generations add hydrodynamic hulls, high-pressure engines, lightweight materials, and optimized propellers.

Two caveats remain: ship lifespans of 20-30 years mean fleet-wide efficiency advances materialize slowly, and size benefits accrue only if vessels sail near full capacity and avoid empty backhauls. Nevertheless, the upscaling trend – container ships jumping from ~3,000 TEU in the 1990s to over 8,000 TEU by the 2010s – has been a net positive for reducing CO_2 per transport unit.

3.4. Operational Practices (Speed, Utilization)

Day-to-day operations strongly influence shipping CO_2 emissions. Speed is the most adjustable "knob," since required propulsive power scales roughly with the cube of speed. After 2008, average container-ship speed fell from ~22-24 kn to ~18-20 kn, cutting fuel burn by 20-30% despite longer voyages – a change later partly reversed when carriers accelerated to meet surging demand. Other operational measures also matter: higher fleet utilization (fewer empty legs) lowers per-cargo emissions; advanced route-planning software accounts for weather and currents; just-in-time arrival reduces idle anchoring; and proactive hull maintenance (removing biofouling) can save 5-10% of fuel.

4. Regulatory Framework

5.1. International Maritime Organization (IMO)

In July 2023, the IMO adopted a new climate strategy aimed at achieving net-zero greenhouse gas emissions by or around 2050. It has also set an objective to cut emissions by 20-30% by 2030 and by 70-80% by 2040, compared to the levels recorded in 2008. (T&E, n.d.)

However, despite these plans and multiple rounds of negotiations, the IMO has yet to implement reduction measures that align the maritime sector with the temperature targets outlined in the Paris Agreement, which aims to limit the global temperature rise this century to well below 2°C compared to pre-industrial levels and to strive to keep it to 1.5°C. (T&E, n.d.)

Furthermore, one of the IMO's earlier major initiatives was the Energy Efficiency Design Index (EEDI), which was approved in July 2011. It represents the first globally binding design standard intended to mitigate climate change caused by shipping. This standard applies to almost all new ships and came into effect in 2013, mandating that new ships improve their energy efficiency, with increasingly stringent standards to be enforced over time. (T&E, n.d.)

It has set different standards for ships to meet depending on their classes and sizes as follows:

1. Phase 1: News ships built between 2015 and 2019 will have to improve their vessel energy efficiency by 10%.
2. Phase 2: Ships built between 2020 and 2024 will have to improve their energy efficiency by 15 and 20% depending on the ship type.
3. Phase 3: Ships delivered after 2025 will have to be 30% more efficient. (T&E, n.d.)

The IMO is also actively pursuing a global sulfur cap on marine fuels as part of its strategy to reduce emissions from the shipping sector and has established the Mediterranean Sea Emission Control Area (ECA) to control emissions of sulfur oxides and particulate matter. (Sinay, 2023)

Moreover, the IMO's greenhouse gas strategy encompasses a range of initiatives including harnessing renewable energy sources (e.g. wind and solar) to power oceangoing, leveraging technological innovations to optimize ship routes, speeds, and operations and enhance fuel efficiency, providing shore power at ports to enable ships to tun off their engines while docked, and fostering sustainability training for maritime professionals. It also emphasizes international collaboration to scale innovation and share best practices across the sector. (Sinay, 2023)

5.2. Global Initiatives

At the global level, several major policy efforts aim to decarbonize the maritime sector. One notable initiative is the Clydebank Declaration, launched at COP26, which promotes the creation of green shipping corridors – zero emissions maritime routes between ports. (Talalasova, 2023)

Additionally, there is a growing global push for investments in infrastructure and human capital, ensuring that the energy production and shipyard capacities and port infrastructure all support the large-scale transition to green technologies, and cleaner operations. These efforts are complemented by financial incentives such as subsidies or tax incentives, support for shipping-specific R&D programs, and the establishment hydrogen hubs near ports to develop zero-emission fuel infrastructure and bridge supply-demand gaps. (Talalasova, 2023)

5.3. Regional Actions

The European Union has introduced several regulatory measures to decarbonize the shipping sector. These include the FuelEU maritime initiative, which sets targets for reducing greenhouse gas intensity of energy used onboard, and the introduction of new subsidies for sustainable fuels as well as penalties for the use of fossil fuels, aiming at encouraging the shift to alternative energy sources. (Swanson, 2023) (Talalasova, 2023)

The European Parliament has also endorsed modifications to the EU Emissions Trading System (EU ETS) that broaden its scope to cover all maritime emissions. This expansion applies to ships exceeding 5,000 gross tonnages operating within the European Economic Area (EEA). Consequently, every company with vessels engaged in EEA trade must submit emission allowances corresponding to a share of their annual greenhouse gas emissions. (Sinay, 2023)

5.4. National Strategies

The United States is driving new investments in green fuels production (hydrogen and ammonia) and in port infrastructure upgrade to support shore power and cleaner fuel use. It is also funding programs to accelerate decarbonization efforts across the maritime sector. (Swanson, 2023)

Moreover, China has implemented a comprehensive set of policies aimed at promoting green and low-carbon development in its shipping industry. These include the "Marine Environment Law of the People's Republic of China," the "Regulations on the Prevention and Control of Marine Environmental Pollution by Ships and Their Related Operations," and the "Green Development Action Plan for the Shipbuilding Industry"(Hu & Dong, 2025)

Additionally, Chinese shipowners are actively engaging in green development through adopting energy-saving technologies and management practices, promoting green shipping initiatives, and enhancing energy efficiency management. (Hu & Dong, 2025)

Chinese ports are also undergoing a green transition, with the application of renewable energy in them and efforts to green port logistics operations. The report "Blue Port Pioneers 2022: Evaluation of Air and Climate Synergy in China's Typical Ports" reveals that by 2021, the shore power coverage rate for specialized berths in China's inland ports reached nearly 100%, significantly curbing air pollutant and greenhouse gas emissions from ships' auxiliary engines while at port. (Hu & Dong, 2025)

5.5. Shipping Companies' Policies

In recent years, some shipping companies have sought to reduce emissions and meet new international pollution regulations by utilizing liquefied natural gas (LNG) to power their ships as an alternative to conventional heavy fuel oil and marine gas oil. (Swanson, 2023) Using LNG can potentially achieve up to a 100% reduction in SOx emissions and up to a 90% decrease in NOx emissions when compared to heavy fuel oil. It can also lead to a reduction of greenhouse gas emissions of up to 10% at best compared to the diesel fuel it replaces, provided the methane leakage remains at an optimistic level. (T&E, n.d.)

Maersk and other shipping companies now view alternative fuels like methanol, ammonia, and hydrogen as the most viable options for a sustainable future. Maersk aims to achieve net-zero carbon emissions by 2040 and is investing billions of dollars in cleaner fuel options, along with other investors. (Swanson, 2023)

5.6. Relation Between Shipping Decarbonization and Sustainable Development Goals

While the UN Sustainable Development Goals (SDGs) were not explicitly designed for the maritime sector, shipping can contribute meaningfully to several of these global objectives. The industry's transition toward low-carbon operations aligns particularly well with four key SDGs: (Clear Seas, 2024)

1. **SDG 7 – Affordable and Clean Energy:** promotes the adoption of renewable and low-emission energy sources, directly supporting the shift away from fossil fuels in maritime transport.

2. **SDG 8 – Decent Work and Economic Growth:** recognizes the importance of sustainable economic development and employment, both of which are supported by the shipping industry through creating millions of jobs worldwide and facilitating global trade.
3. **SDG 13 – Climate Action:** calls for urgent measures to combat climate change and its impacts by regulating emissions and promoting renewable energy, which are goals directly relevant to reducing CO_2 emissions from shipping.
4. **SDG 14 – Life Below Water:** aims for the protection of marine ecosystems, which are affected by pollutants and emissions from shipping, so greening maritime transport can mitigate these impacts.

5. Data Sources, Analysis and Kaya Decomposition

Understanding the contribution of large merchandise ships to global CO_2 emissions requires decomposing the aggregate emissions into more granular factors. To this end, we apply a Kaya-type identity tailored to the maritime sector, allowing us to disentangle the different economic and technological components that shape emissions trends.

5.1. Rationale for Using the Kaya Decomposition

The Kaya identity is a widely accepted framework in climate economics that expresses CO_2 emissions as the product of key driving forces: economic activity, transportation intensity, energy intensity, and carbon intensity. In the context of maritime shipping, we adapt this identity to reflect the sector-specific characteristics of freight transport. Our decomposition follows the form:

$$CO_2 = GDP \times \left(\frac{Tonnes \times Miles}{GDP}\right) \times \left(\frac{Fuel\ Consumption}{Tonnes \times Miles}\right) \times \left(\frac{CO_2}{Fuel\ Consumption}\right)$$

Each term in this equation highlights a different aspect:

- **Transportation Intensity** = Tonnes × Miles / GDP
- **Fuel Intensity** = Fuel Consumption / (Tonnes × Miles)
- **Carbon Intensity** = CO_2 / Fuel Consumption

By decomposing CO_2 emissions into these components, we can trace the historical evolution of shipping emissions, analyse their drivers, and explore potential decoupling between economic growth and environmental impact.

5.2. Description of Data Sources

The decomposition relies on four main datasets, each corresponding to a term in the equation above. All data sources are from reputable institutions and were chosen for their transparency, coverage, and sectoral relevance.

- **CO_2 Emissions – EDGAR**

We use the *Emissions Database for Global Atmospheric Research (EDGAR)* to obtain annual CO_2 emissions from maritime transport. EDGAR provides consistent global emissions data for the IPCC sector 1A3d(ii) "Water-borne navigation" across countries and years.

- **GDP – OECD Data Explorer**

To normalize transport volumes and measure economic activity, we rely on GDP data (current prices, USD) from the *OECD Data Explorer*. This allows us to compute transportation intensity as freight activity per dollar of economic output.

- **Tonnes × Miles – DNV.GL Maritime Outlook**

Freight activity is approximated using data from *DNV.GL's Maritime Forecast to 2050*. This report includes long-term projections and historical values for global tonne-miles, which we interpret as a proxy for shipping output.

- **Fuel Consumption – IEA Marine Bunker Deliveries**

Fuel usage is estimated using data from the *International Energy Agency (IEA)*, specifically deliveries of oil products to international marine bunkers. While this is an indirect proxy, it remains one of the best publicly available sources for maritime fuel consumption across regions and time.

5.3. Historical Evolution of Maritime CO_2 Drivers (1980 – 2019)

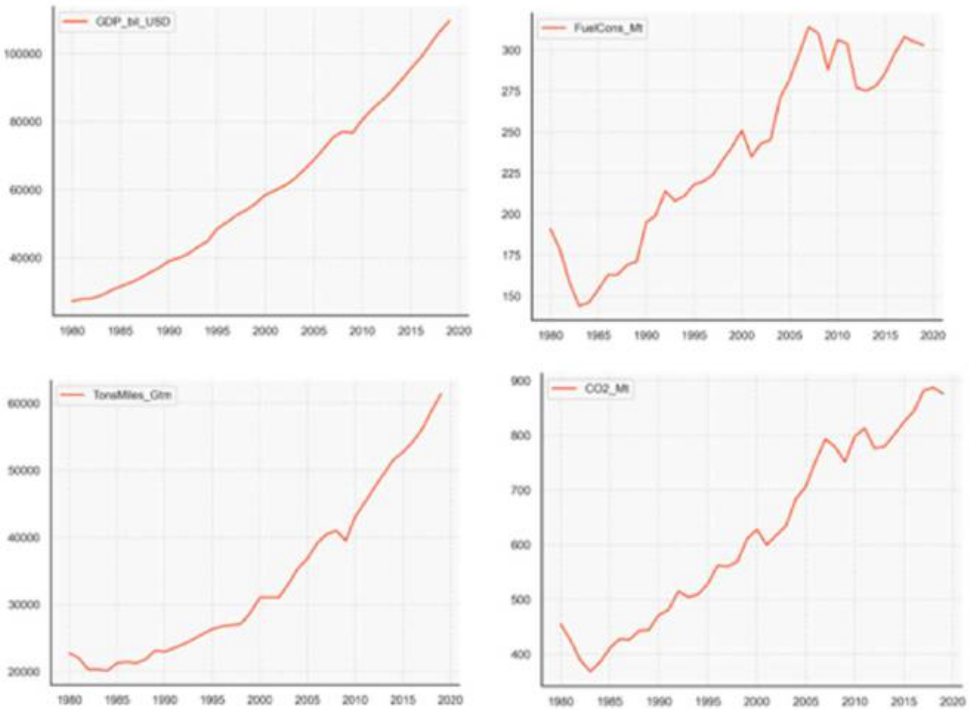

Figure 1: Historical Evolution of Maritime CO2 Drivers (1980-2019)

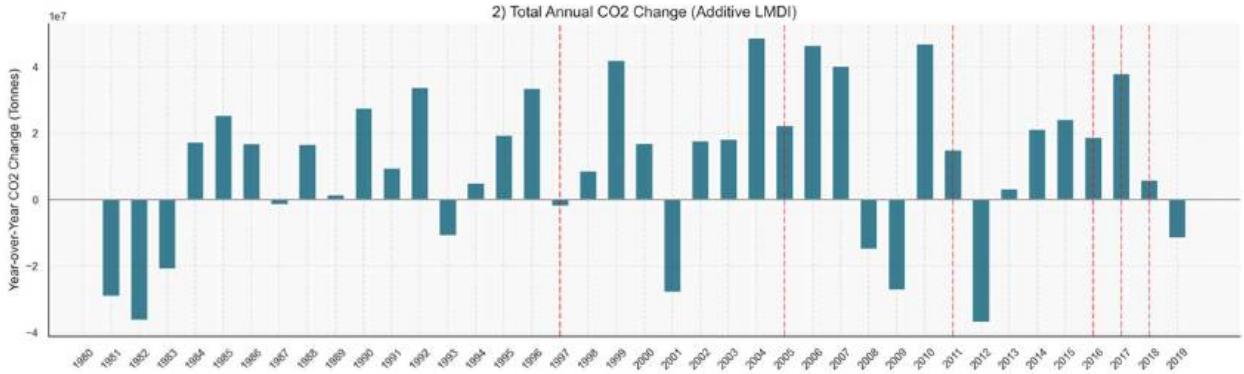

Figure 2: Total Yarnover Change in Maritime CO_2, calculated with LMDI

Figure 1 contains four-line plots.

• **Panel A (upper left)** tracks global GDP, expressed in *billion US dollars* from 1980 to 2019. The curve rises almost exponentially – from roughly 28 000 bn USD in 1980 to more than 105 000 bn USD by 2019 – with only a brief slowdown during the 2008-09 financial crises.

• **Panel B (upper right)** depicts marine fuel consumption, measured in *megatons*. After an early1980s trough, the series doubles over the study period, peaking close to 310 Mt in 2008 and oscillating thereafter – evidence of both efficiency gains and demand fluctuations.

• **Panel C (lower left)** displays tonne miles (gigatonne miles). Activity remains near 20 000 Gtm until the late 1980s, accelerates after 2000, and surpasses 60 000 Gtm by 2019, with noticeable dips in 2009 and 2011.

• **Panel D (lower right)** shows maritime CO_2 emissions in *megatons*. The series dips in the early 1980s, then climbs steadily to about 890 Mt in 2018. Short plateaus appear around 2001, 2009, and 2013, mirroring macroeconomic contractions.

Figure 2 is a bar chart that reports the total yarnover change in maritime CO_2, calculated with an additive Logarithmic Mean Divisia Index (LMDI). Bars above the zero line indicate years when aggregate emissions rose; bars below zero signify reductions. Large positive jumps dominate the early2000s, reaching more than $4 \times 1074 \times 10^7\ t$, whereas negative spikes in 1983, 2009, and 2015 coincide with recessions or efficiency breakthroughs. Vertical red dashed lines highlight specific benchmark years.

5.4. Maritime Efficiency and Intensity Indicators with Key Policy Landmarks

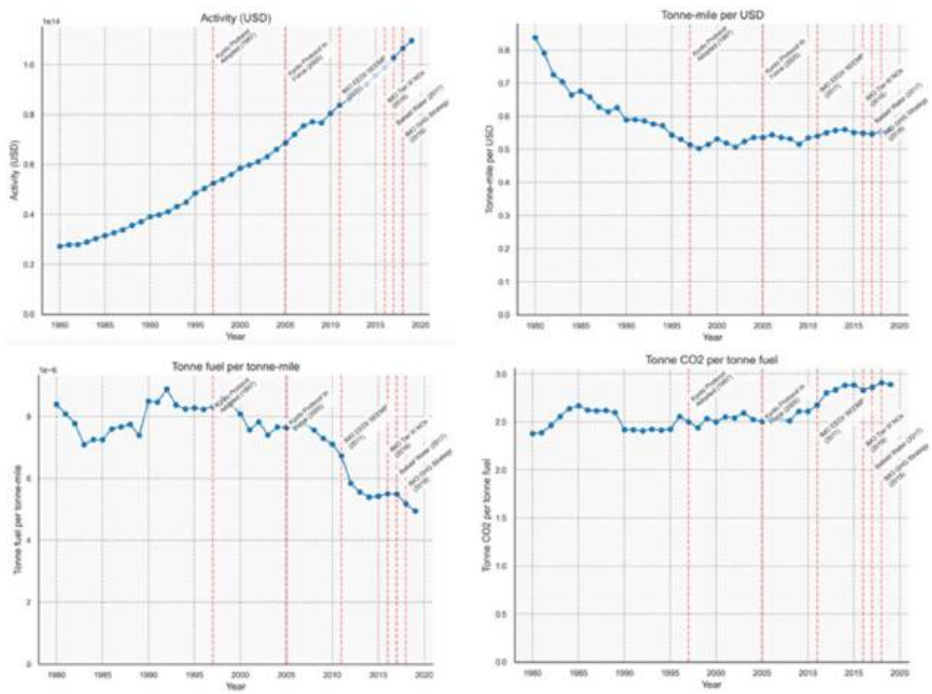

Figure 3: Historical Trends of the Four Components of the Kaya Decomposition

• **Panel 1 (upper left) s**hows global shipping activity (proxied by GDP) in constant USD from 1980 to 2019, using scientific notation ($\times 10^{14}$ USD). The trend is near-exponential, rising from ~2.7×10^{13} USD in 1980 to over 1.1×10^{14} USD in 2019, with a slight dip during the 2008–09 crisis. Red dashed lines mark major climate policy milestones (e.g., Kyoto Protocol, IMO Strategy), but no clear shifts follow these, highlighting macroeconomic inertia.

• **Panel 2 (upper right)** tracks transportation intensity defined by ton-miles per USD. After a sharp drop from 0.84 to 0.68 in the early 1980s, values declined more gradually to ~0.50 by 1998. Post-2000, the trend stabilizes and slightly rises, reaching ~0.56 in 2019. The initial decline reflects reduced transport intensity during early globalization, while the later plateau indicates limited further decoupling.

• **Panel 3 (lower left)** shows fuel intensity defined by tonne fuel per tonne mile ($\times 10^{-6}$). After a high plateau (~8.4) in 1980, values decline with peaks in the early 1990s and 2001. A sharper drop follows 2010, falling below 6×10^{-6} by 2019. This aligns with the 2011 adoption of IMO's EEDI and SEEMP, suggesting these measures effectively reduced fuel use per ton mile.

• **Panel 4 (lower right)** shows carbon intensity defined by tonne CO_2 per tonne fuel, mostly stable between 2.4 and 2.9 t CO_2/t fuel. Aside from a brief dip in the late 1990s, values remain flat, with a slight rise post-2010. This indicates minimal adoption of low-carbon fuels, implying emissions cuts rely more on transport or fuel efficiency than on fuel decarbonization.

The four panels show partial, uneven decoupling of maritime emissions drivers. Economic activity rises steadily; transportation intensity declines early but then levels off; fuel intensity improves notably after 2010, likely due to IMO regulations; carbon intensity remains flat. Overall, policies have boosted efficiency (Panel 3) but failed to alter the fuel mix (Panel 4).

5.5. Summary of Key Drivers, 1980 vs 2019

Table 1: Summary of Key Drivers 1980 vs 2019

	1980	2019	Growth Factor	Growth Rate
GDP (Trillion USD)	27.18	109.69	4.04	+304%
Transport intensity (tonne-mile per USD)	0.84	0.56	0.67	-33%
Fuel intensity (g fuel per ton-mile)	8.39	4.94	0.59	-41%
Carbon intensity (kg CO2 per tonne fuel)	2,378.58	2,889.69	1.21	+21%
CO2 emissions (Mtonnes)	454.31	875.58	1.93	+93%

The table compares baseline (1980) and end period (2019) values for the four Kaya identity components and the resulting maritime CO_2 emissions. Between 1980 and 2019, global economic output quadrupled, while maritime CO_2 emissions rose by only 93%. This partial decoupling is attributable to substantial improvements in **transportation intensity** (–33%) and **fuel intensity** (–41%), which together offset much of the activity growth. In contrast, **carbon intensity** registered a 21% increase, indicating a fuel mix that became slightly more carbon dense or a limited uptake of low carbon alternatives. Overall, the figures confirm that efficiency gains slowed the rise in absolute emissions driven by expansive global trade.

5.6. Cumulative Waterfall Decomposition of CO2

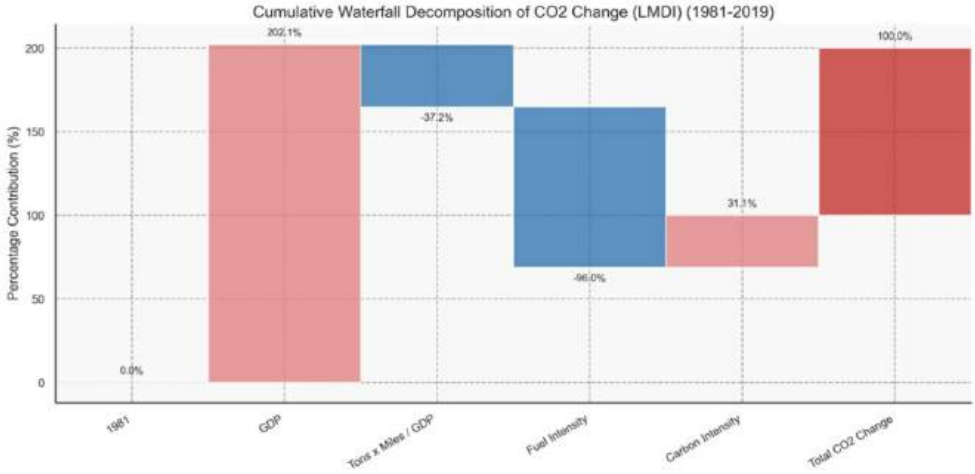

Figure 4: Cumulative Waterfall Decomposition of CO2 (LMDI)

The graphic is a waterfall chart that allocates the cumulative percentage change in maritime CO_2 emissions over 1981-2019 to the four Kaya identity drivers, using the additive (LMDI) method.

GDP adds **+202%**, showing that economic expansion alone would have tripled emissions.

Transport intensity contributes **–37%**, indicating that reduced ton-miles per dollar of output partially offsets GDP growth.

Fuel intensity delivers the largest mitigating effect at **–96%**, reflecting improvements in vessel efficiency and operational practices.

Carbon intensity adds back about **+31 %**, revealing the counteracting influence of a marginally more carbon dense fuel mix (or limited uptake of low carbon fuels).

Overall, the waterfall confirms the results stated in the previous section that technological and operational efficiencies, not fuel decarbonization, have been the dominant brakes on emission growth during the period studied.

5.7. Compound Annual Growth Rates of Kaya Identity Drivers

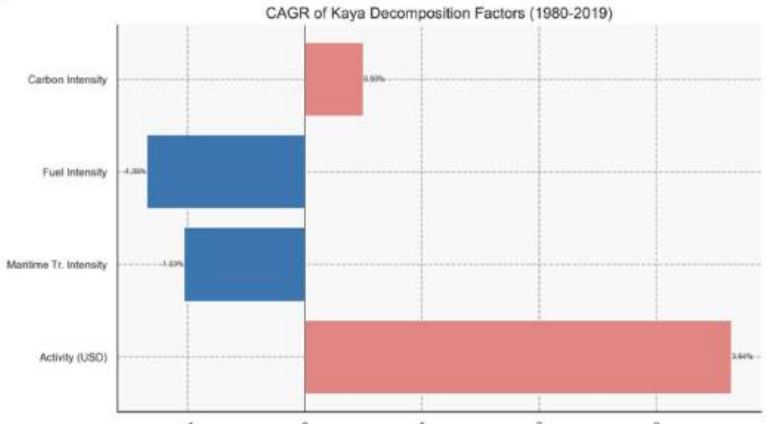

Figure 5: Compound Annual Growth Rates of Kaya Identity Drivers

This bar chart shows the **compound annual growth rate (CAGR)** of the four emissions drivers. Economic activity grows strongly at +3.64%/yr, driving emissions upward. Efficiency improves, with transport intensity falling −1.03%/yr and fuel intensity −1.35%/yr, partially offsetting growth. Carbon intensity rises slightly (+0.50%/yr), reflecting minimal fuel mix decarbonization. Overall, efficiency gains reduce growth pressure by ~2 points, but further emission cuts require cleaner fuels or lower demand.

5.8. Normalized Co-Movement of Fuel and Transport Intensities

This dual axis line chart presents **Z-scored** time series for **fuel intensity** (blue) and **maritime transport intensity** (red). By standardizing each series to zero mean and unit variance, the plot highlights relative departures from long run averages rather than absolute magnitudes.

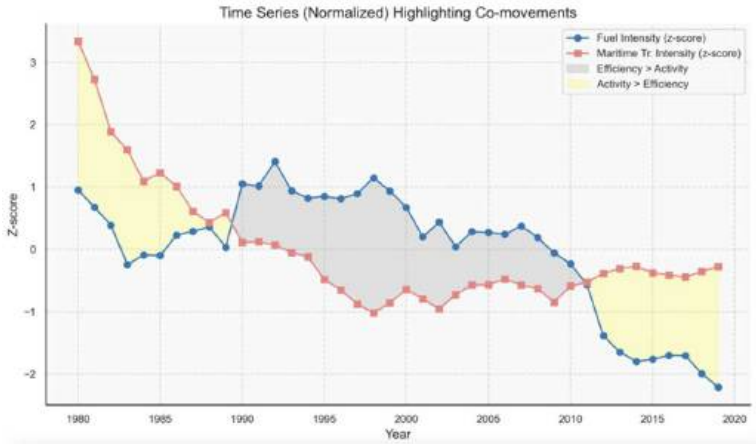

Figure 6: Normalized Co-Movement of Fuel and Transport Intensities

In the early 1980s (yellow), transport intensity deviated more from its mean than fuel intensity, reflecting rapid globalization. Between 1990 and 2010 (grey), fuel efficiency gains led, driven by technical improvements. After 2012 (yellow), transport demand rose while fuel efficiency stalled. These shifts illustrate a tug-of-war between technological progress and trade-driven emissions.

5.9. Relationship between Fuel Intensity and Maritime Transport Intensity

The scatter plot shows the relationship between fuel intensity and maritime transport intensity. Each blue marker represents a single year between 1980 and 2019. A quadratic polynomial (red curve) is fitted to the data. Its coefficient of determination, **$R^2 = 0.03$**, signals an almost negligible explanatory power.

The flat, low R^2 profile confirms that long-run variations in vessel energy efficiency are largely orthogonal to changes in trade intensity, implying that the two levers can, and must, be tackled independently in policy design.

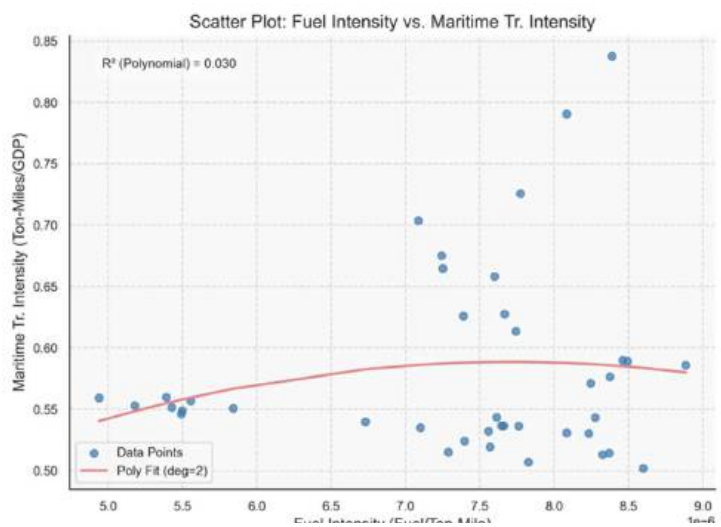

Figure 7: Graph showing the Relationship between Fuel and Transport Intensities

5.10. Comparison with the Interpolated Data

Data from 2020 to 2024 were estimated using XGBoost regression models, trained on available historical trends.

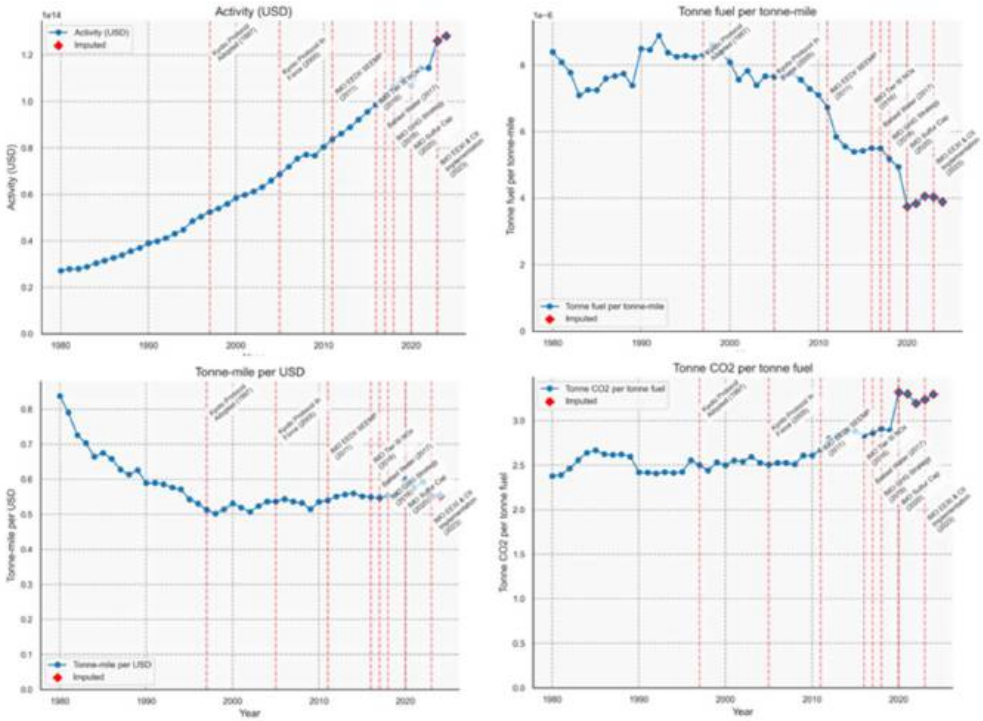

Figure 8: Graph showing the Historical Data with the Interpolated Data for the Kaya Decomposition Components

- **Activity** keeps accelerating, consistent with past behavior but is sensitive to macroeconomic uncertainty.
- **Fuel intensity** slope steepens markedly; achieving a 1.1×10^{-6} absolute cut in four years implies rapid fleet retrofits and full compliance with EEXI/CII – ambitious but aligned with recent IMO regulation.
- **Transport intensity** shows no post 2019 improvement; history suggests limited potential left for further structural decoupling.
- **Carbon intensity** forecast contradicts expected fuel switching to LNG, methanol and biofuels; a continued rise is unlikely once alternative fuels scale.

Reliability Comment: The imputed extension credibly reflects ongoing GDP growth and regulatory driven efficiency gains, yet likely overstates future carbon intensity and the pace of fuel intensity improvements. Because the Boost model is trained on past correlations, it cannot fully capture forthcoming step changes in fuel technology or demand shocks. Confidence is therefore moderate for GDP and transport intensity, lower for fuel and carbon intensity projections.

5.11. Summary of Key Drivers 1098 vs 2024

Table 2: Summary of Key Drivers 1098 vs 2024

	1980	2024	Growth Factor	Growth Rate
GDP (Trillion USD)	27.18	128.25	4.72	+372%
Transport intensity (tonne-mile per USD)	0.84	0.55	0.66	-34%
Fuel intensity (g fuel per ton-mile)	8.39	3.89	0.46	-54%
Carbon intensity (kg CO2 per tonne fuel)	2,378.58	3,295.89	1.39	+39%
CO2 emissions (Mtonnes)	454.31	906.89	2.00	+100%

GDP climbs more quickly in the forecast, sustaining pressure on emissions. Transport intensity shows no further reduction, suggesting that structural trade shifts have reached diminishing returns. Fuel intensity improvement accelerates, delivering the bulk of additional mitigation. As for the carbon intensity, a projected jump in it is at odds with imminent adoption of LNG, methanol, and biofuels, suggesting that the model overfits recent HSFO rebound.

As a result, CO_2 grows only 4 Mt yr^{-1} on average post 2019, an historic low, but outcome is fragile, hinging on optimistic fuel efficiency gains and pessimistic fuel mix assumptions.

GDP and transport intensity projections appear reasonable, fuel intensity improvement is ambitious, yet policy aligned, and carbon intensity estimate likely overstates emissions, thereby rendering the aggregate CO_2 forecast conservative.

5.12. Compound Annual Growth Rates of Kaya Identity Drivers

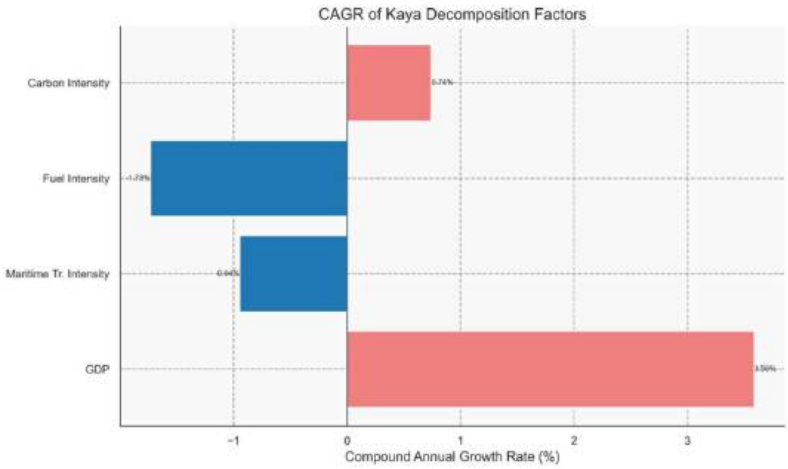

Figure 9: Compound Annual Growth Rates of Kaya Identity Drivers with the Interpolated Data

The historical pattern shows economic growth outpacing the combined mitigating effects of declining transport and fuel intensity, with a modest rise in carbon intensity further eroding gains. The forecast to 2024 reinforces this structure but with two notable adjustments: (i) fuel intensity declines faster, suggesting strong compliance with forthcoming technical measures; and (ii) carbon intensity rises even more quickly, counterintuitive to policy targets for alternative fuels. Therefore, efficiency gains and fuel carbonization nearly cancel, leaving aggregate emissions growth barely lower than the historical pace.

5.13. Normalized Co-Movement of Fuel and Transport Intensity

In the imputed extension, fuel intensity remains deeply negative, but no new efficiency surge appears; transport intensity shows a mild uptick, keeping the *activity* > *efficiency* regime intact through 2024.

Despite the earlier steep numeric drop in fuel intensity, its Z-score stabilizes, meaning the model foresees no additional shock relative to the already low 2019 baseline. Because transport intensity does not fall further, the yellow "activity driven" configuration continues.

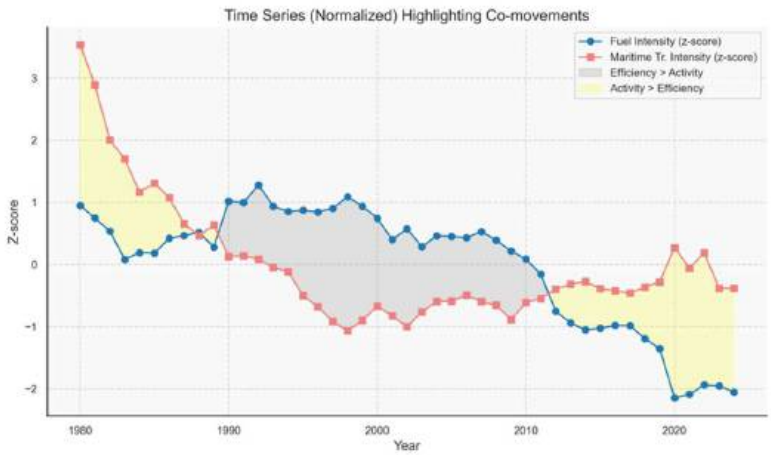

Figure 10: Normalized Co-Movement of Fuel and Transport Intensity with the Interpolated Data

5.14. Relationship between Fuel Intensity and Maritime Transport Intensity

Historically, fuel efficiency and trade intensity exhibited almost no statistical coupling. The 2020-2024 extension weakens that already tenuous linkage (R^2 halves). This occurs because the machine learning forecast pushes fuel intensity sharply lower without a corresponding change in transport intensity, thereby stretching the cloud leftward but not upward. The result reinforces the inference that technological efficiency gains and structural trade dynamics are essentially independent

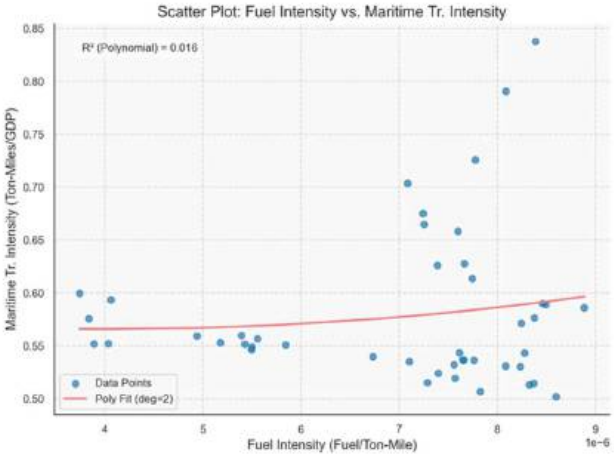

Figure 11: Graph showing the Relationship between Fuel and Transport Intensities with the Interpolated Data

levers – an insight that remains valid, and even clearer, after imputation.

The new low fuel intensity points are plausible given impending EEXI/CII standards, yet their independence from trade intensity may be exaggerated: operational slow steaming (a likely compliance strategy) could reduce ton-miles as well as fuel use, tightening the correlation. Consequently, the vanishing R^2 should be viewed cautiously; Real World behavioral adjustments may restore some degree of relationship between the two variables.

5.15. Assumptions and Limitations

Several caveats must be acknowledged in this analysis:

- **Fuel Consumption as a Proxy:** We use bunker fuel deliveries as a proxy for actual fuel burned. This may overestimate or underestimate consumption depending on inventory dynamics and data coverage.
- **CO_2 Aggregation:** EDGAR data aggregates all waterborne navigation emissions, not solely those from large merchandise ships, which introduces potential bias.
- **Data Coverage:** There are missing or outdated values for some years, especially post-2019, which may affect long-term trend analysis.
- **Global Averages:** Using global values smooths out regional heterogeneities in regulation, fleet composition, or trade patterns.

6. Forecasts: CO_2 Emission Scenarios for Global Merchandise Shipping

Figure 12 presented illustrates various projected trajectories of CO_2 emissions from global commercial shipping between 2024 and 2055. It serves as a visual summary of plausible emission pathways based on different policy and economic assumptions. Each scenario reflects distinct drivers of emissions identified earlier in this paper, ranging from trade growth and energy efficiency to regulatory and geopolitical dynamics.

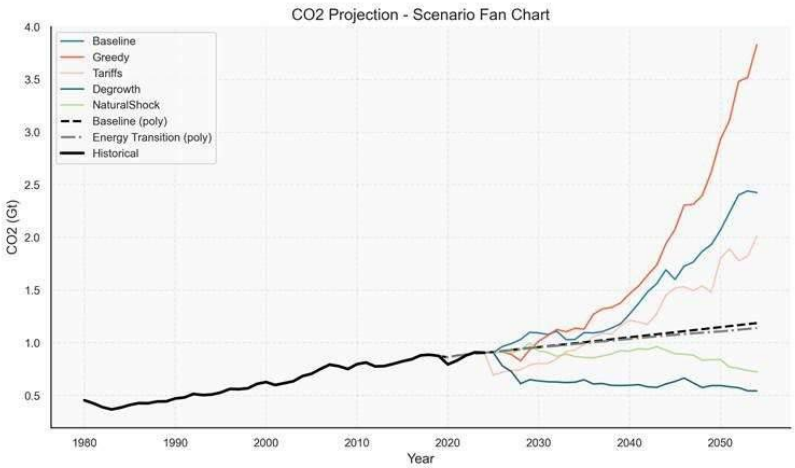

Figure 12: CO2 Projections in Different Scenarios

The **Baseline** scenario projects a sustained increase in emissions, culminating in a rise of approximately 167.5% by 2050 relative

to 2023 levels. This scenario assumes a continuation of existing trends without significant policy interventions or technological breakthroughs.

The **Greedy** scenario exhibits the steepest upward trend, with emissions more than tripling by 2050. This reflects a global environment prioritizing economic expansion and liberalized trade, with minimal regard for environmental constraints.

In contrast, the **Tariffs** scenario presents a moderated but still rising emissions path. Despite a contraction in global trade activity due to protectionist policies, overall emissions increase by over 120% highlighting inefficiencies in production and logistics and emphasizing that trade reductions alone are insufficient for emission mitigation.

The **Degrowth** scenario presents the most pronounced decline in emissions, achieving a reduction of over 40% relative to 2024. It models a strategic contraction of global economic output with accompanying reductions in transport demand and energy intensity, aligning with emerging literature advocating degrowth as a pathway to sustainability.

The **Natural Shock** scenario reflects emissions declines of approximately 20%, driven by climate-related disruptions that impair economic and logistic systems. While effective in the short-term, the scenario assumes no structural decarbonization, and emissions stabilize rather than decline further.

The **Baseline (poly)** and **Energy Transition (poly)** lines are polynomial projections based on OECD's GDP forecasts and historical CO_2-GDP relationships. Using regression analysis, we estimated future CO_2 emissions under two economic trajectories: conventional growth and transition-oriented reform. The Baseline (poly) reflects emissions rising in line with GDP, while the Energy Transition (poly) shows partial decoupling. These curves translate macroeconomic outlooks into sector-specific emissions forecasts.

Finally, the **Historical** series establishes the empirical foundation of the model, showing gradual but persistent emissions growth up to 2023. The divergence in post-2023 projections encapsulates the range of future outcomes contingent upon global economic choices, policy decisions, and technological advancements in the maritime sector.

7. Conclusion

Our analysis of CO_2 emissions from large merchandise ships, grounded in a Kaya-type decomposition, provides valuable insight into the interplay between economic growth, transport demand, energy use, and carbon intensity in the maritime sector.

In this report, we distinguish between historical and imputed datasets. Historical data refers to observed and recorded values sourced directly from official databases, including EDGAR (for CO_2 emissions), OECD (GDP), DNV.GL (tonne-miles), and the IEA (fuel consumption). These

values are limited to periods where verified data is publicly available. In contrast, the imputed data includes recent years where gaps in historical records exist. These missing values were estimated using XGBoost regression models, trained on available historical trends. While historical data ensures empirical accuracy, imputed data allows us to extend the analysis to more recent years, providing a predictive perspective, albeit with a degree of model-based uncertainty.

The decomposition reveals several key findings. Transport demand (tonne-miles) has increased, but not proportionally to GDP growth, indicating increasing economic efficiency in trade logistics. Fuel intensity shows only partial improvements, pointing to slow technological diffusion or offsetting effects from larger ships or increased distances. Carbon intensity remains stagnant, underscoring the ongoing reliance on fossil fuels in maritime propulsion systems. Overall, despite rising freight activity, total emissions have not risen significantly, suggesting a relative decoupling between emissions and trade.

Our study is subject to several limitations. First, there are temporal gaps in the data, especially for the years after 2019. Second, the use of aggregated data across ship types and regions reduces the precision of our findings. Third, the reliance on proxy variables (e.g., bunker fuel deliveries) introduces a degree of uncertainty. Lastly, there were inconsistencies in data definitions across sources (e.g., bunker fuels vs. onboard fuel use).

Several recommendations can be considered to improve the precision and policy relevance of this work. Future research could use disaggregated datasets, distinguishing by ship type (container, bulk, tanker) and incorporate AIS data or satellite-tracked maritime flows to refine tonne-mile estimates. They could also extend the decomposition with a regional breakdown, to better capture the effects of localized climate policies or port regulations. Finally, they can explore alternative fuels and technological scenarios, evaluating their impact on future carbon intensity.

This study highlights the importance of considering sectoral dynamics in climate mitigation strategies. While absolute decoupling has not yet occurred, the reduction in transportation intensity offers a pathway for sustainable growth. The persistence of high carbon intensity suggests a clear policy gap in which greater regulation and investment in green maritime technologies will be required to meet the sustainable development goals. Ultimately, the shipping sector remains a critical frontier for achieving the twin goals of economic development and climate sustainability.

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Appendix

Mathematical Formulation of Scenarios' Construction

In constructing our CO_2 emission scenarios, we begin by breaking total annual shipping emissions into four fundamental components. The first is **economic activity**, which measures the value of goods shipped in constant dollars. The second is **transport intensity**, capturing how much cargo is moved per dollar of goods. The third is **fuel intensity**, reflecting how much bunker fuel is consumed for each unit of transport work. Finally, **carbon intensity** describes how much CO_2 is released when a given quantity of fuel is burned. Multiplying these four elements together yields an estimate of total emissions; changes in any component naturally influence the overall CO_2 outcome.

To project each component into the future, we analyse its recent behaviour over a fixed historical window. We calculate year-to-year percentage changes in each factor and summarize those changes by taking the median growth rate (our "drift") and the dispersion around that median (our "volatility"). These two summary statistics capture the typical pace of change and the degree of fluctuation. We then model future annual changes by combining the historical drift with a random "shock" drawn from a fat-tailed probability distribution. This choice of distribution allows for occasional large deviations—reflecting unexpected events—while ensuring that factor values remain non-negative.

Starting from the last observed year, we update each component forward one year at a time. In each step, we apply the historical drift, add a scaled random shock to capture uncertainty, and adjust the result so it never falls below zero. After these stochastic updates, we introduce scenario-specific adjustments: for example, in a "Tariffs" scenario we reduce economic activity, or in a "Greedy" scenario we increase fuel intensity. Once all components have been updated and any scenario shocks applied, we recombine them to get that year's CO_2 estimate. Repeating this process for many years—often with multiple Monte Carlo draws—produces the range of possible emission trajectories shown in the fan chart. This methodology thus integrates empirical trends, random variability, and explicit policy or economic interventions into a coherent framework for exploring future shipping emissions.

Remark:

ChatGPT has been leveraged as a tool in the following activities:

1. To support us throughout the research.
2. To format the text in a more academic style.
3. To help us reduce the size of some paragraphs to meet overall size constraints.